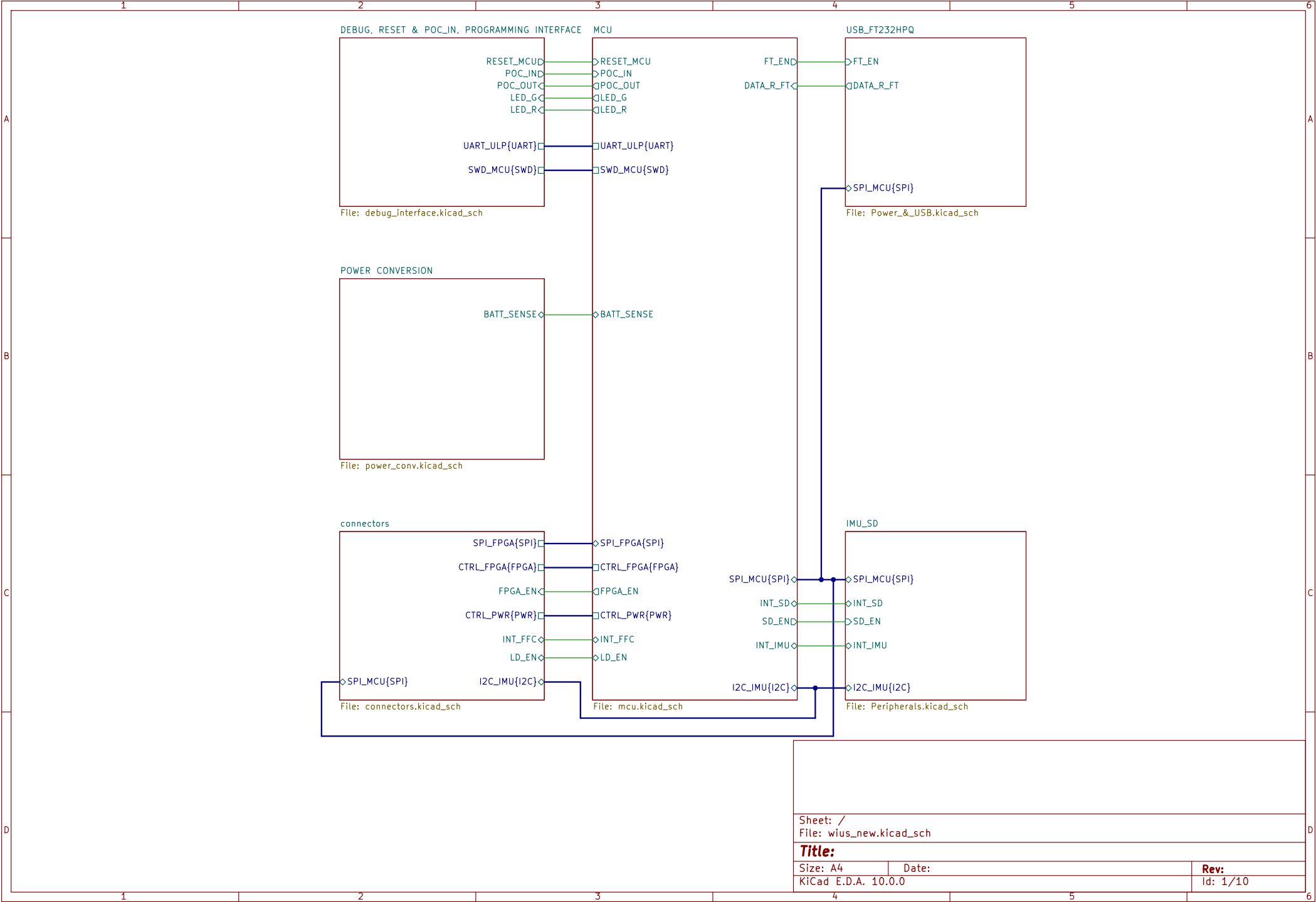
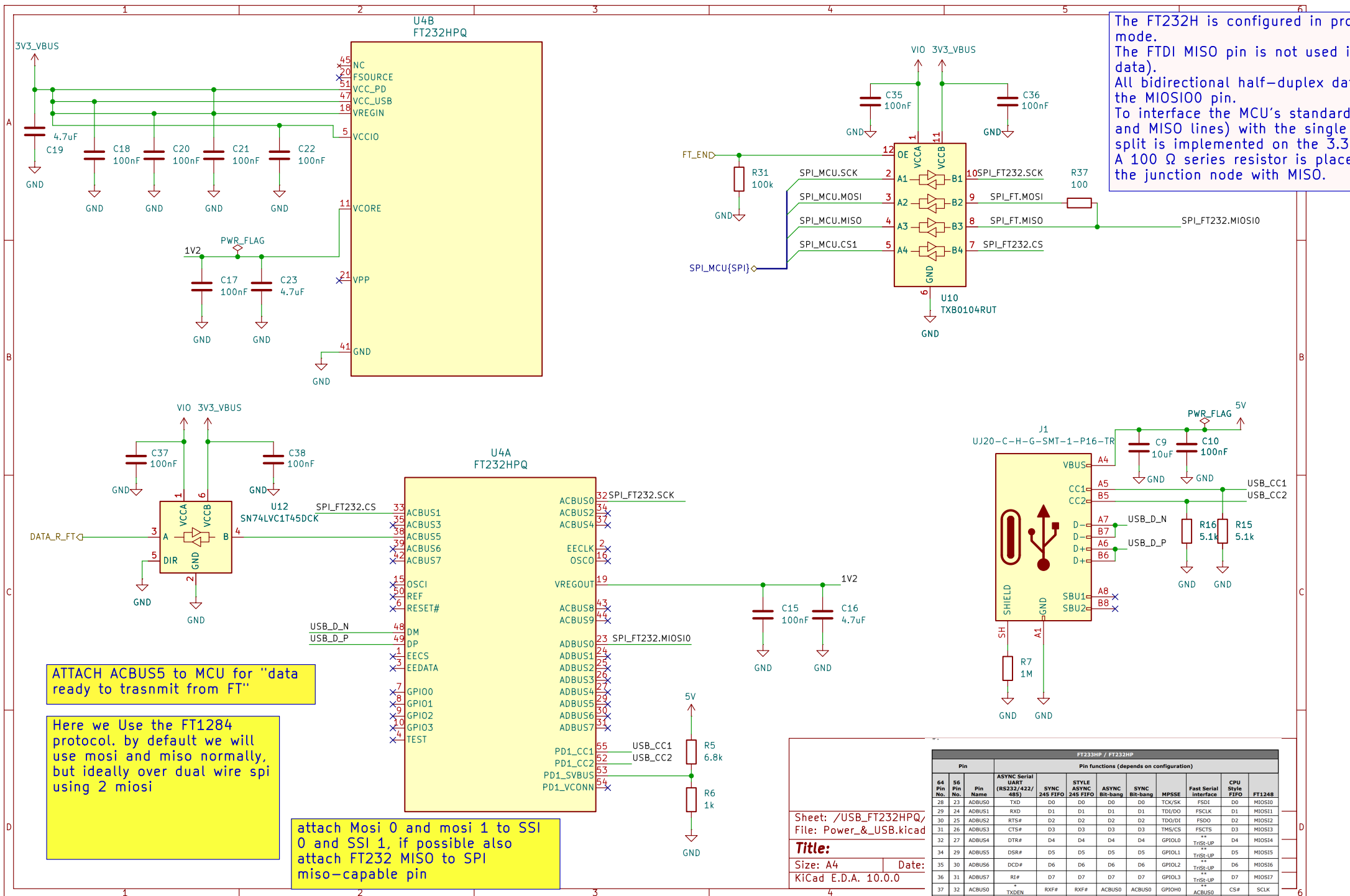
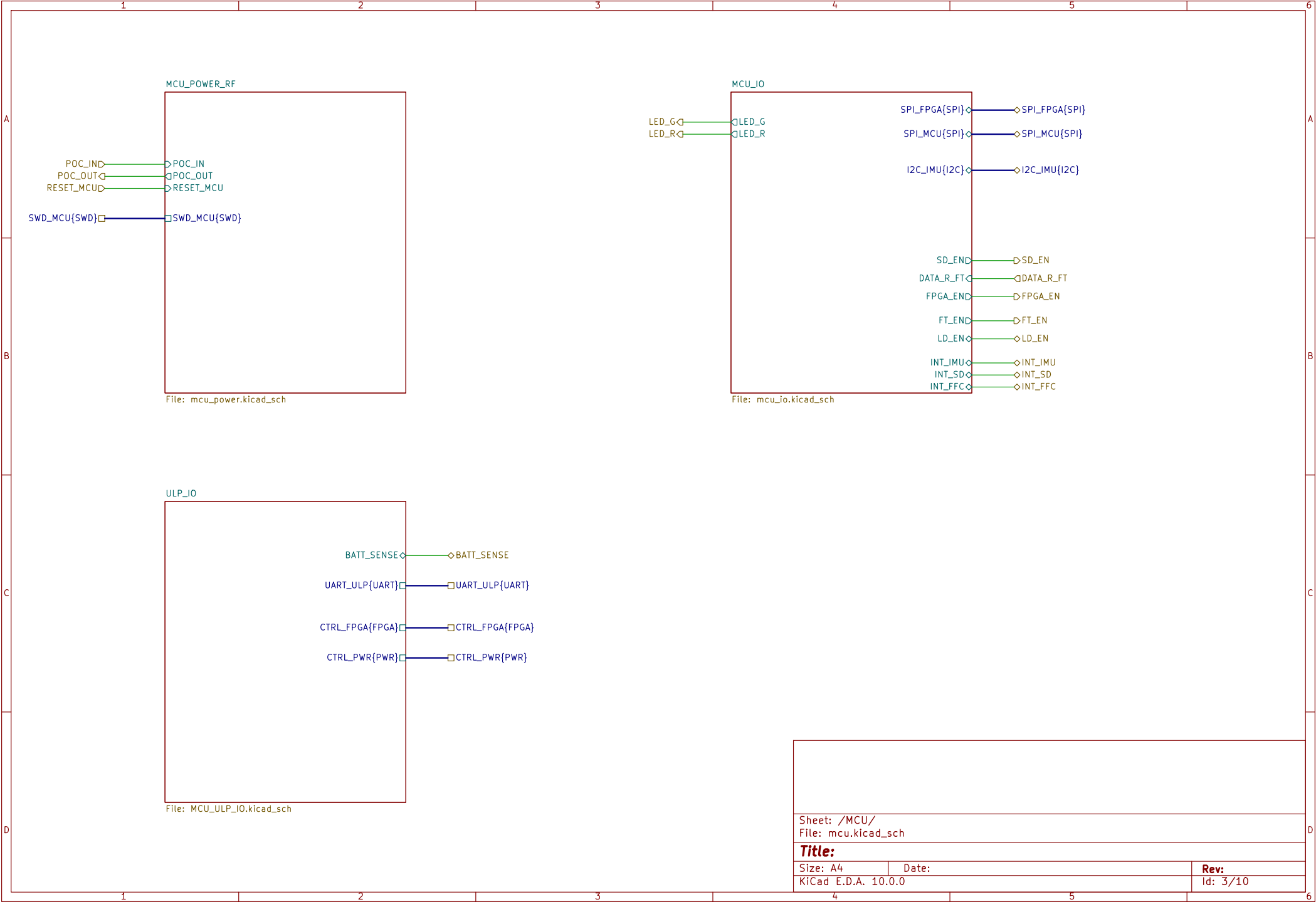








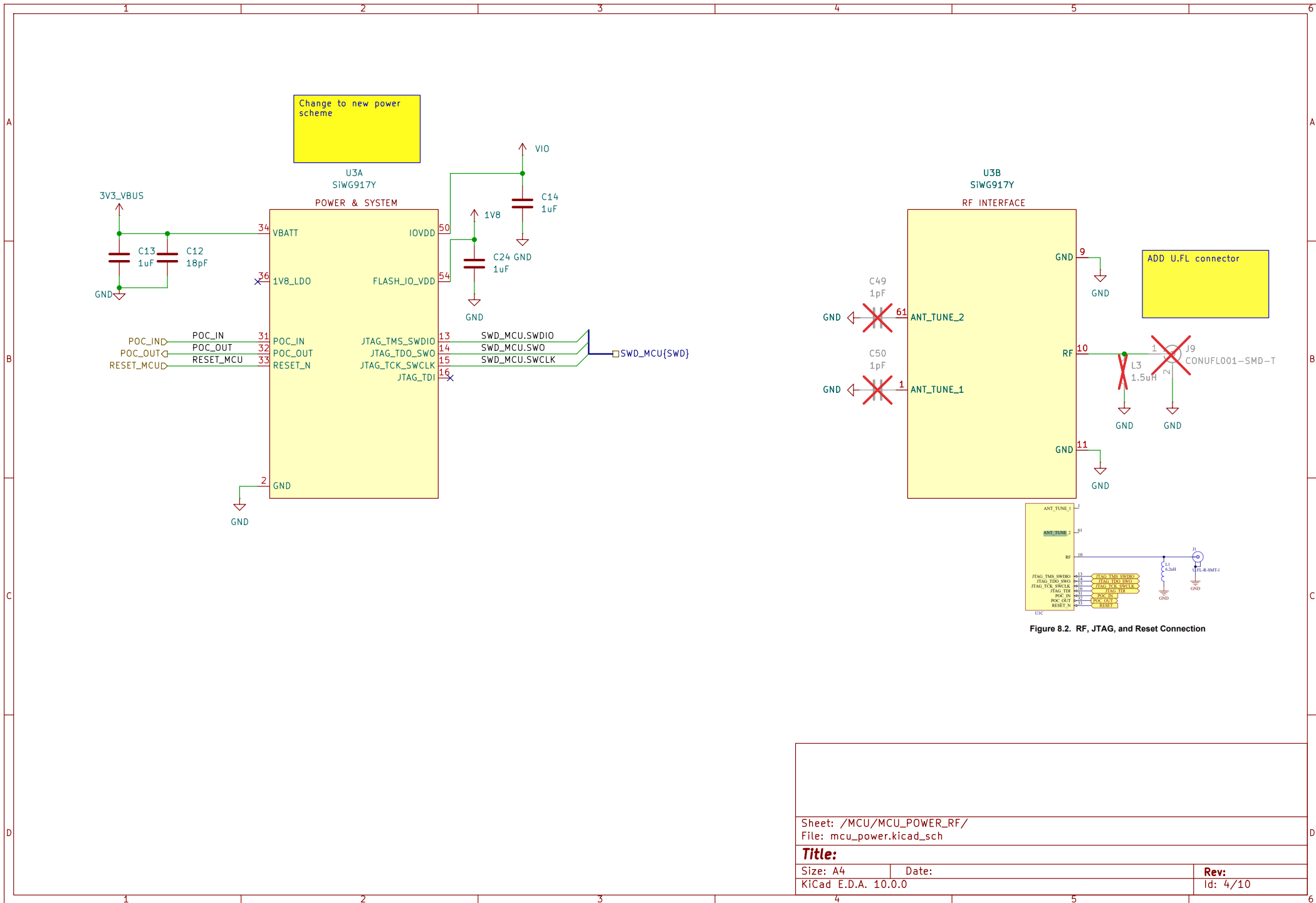


Figure 8.2. RF, JTAG, and Reset Connection

Sheet: /MCU/MCU_POWER_RF/
File: mcu_power.kicad_sch

Title:

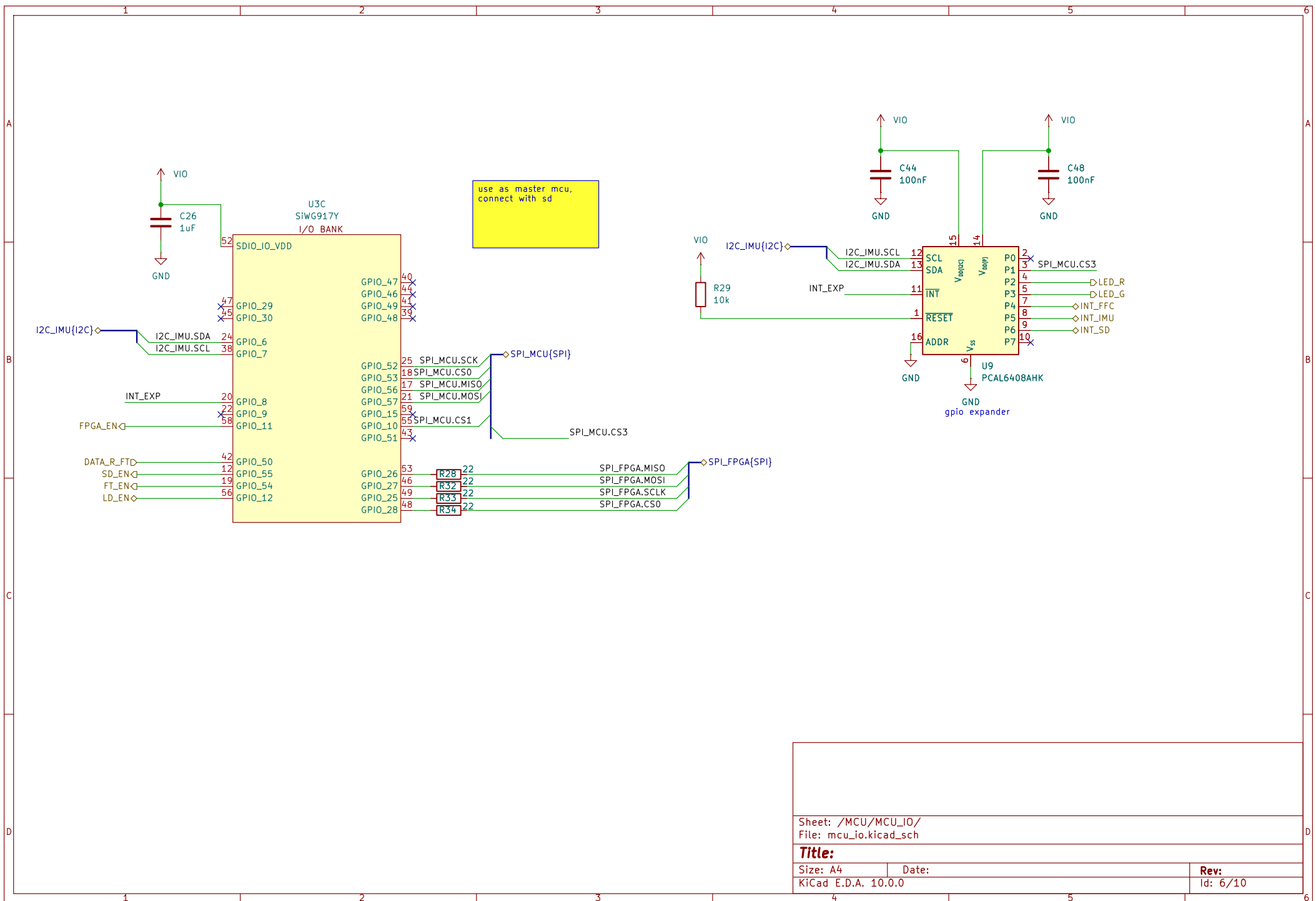
Size: A4

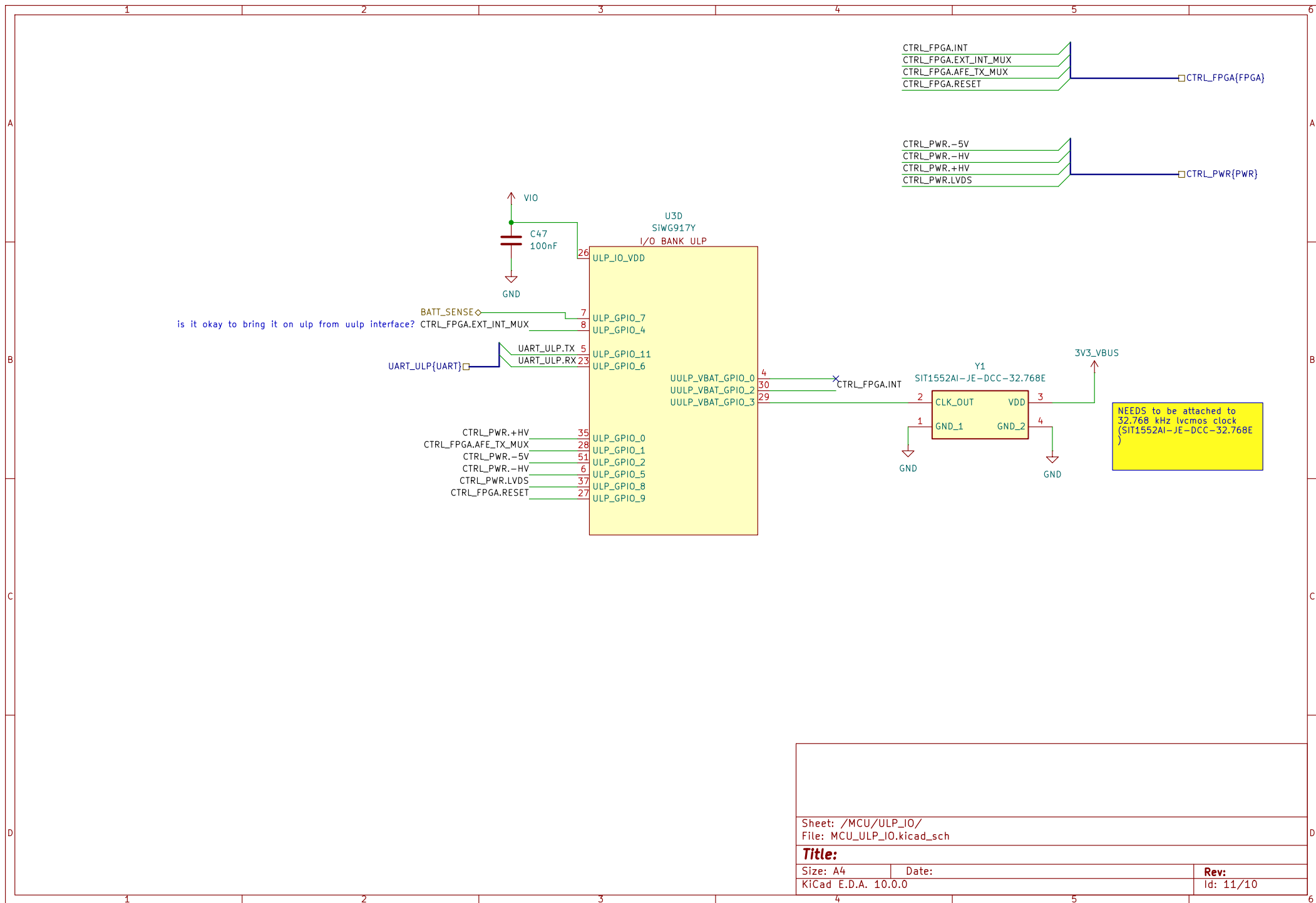
Date:

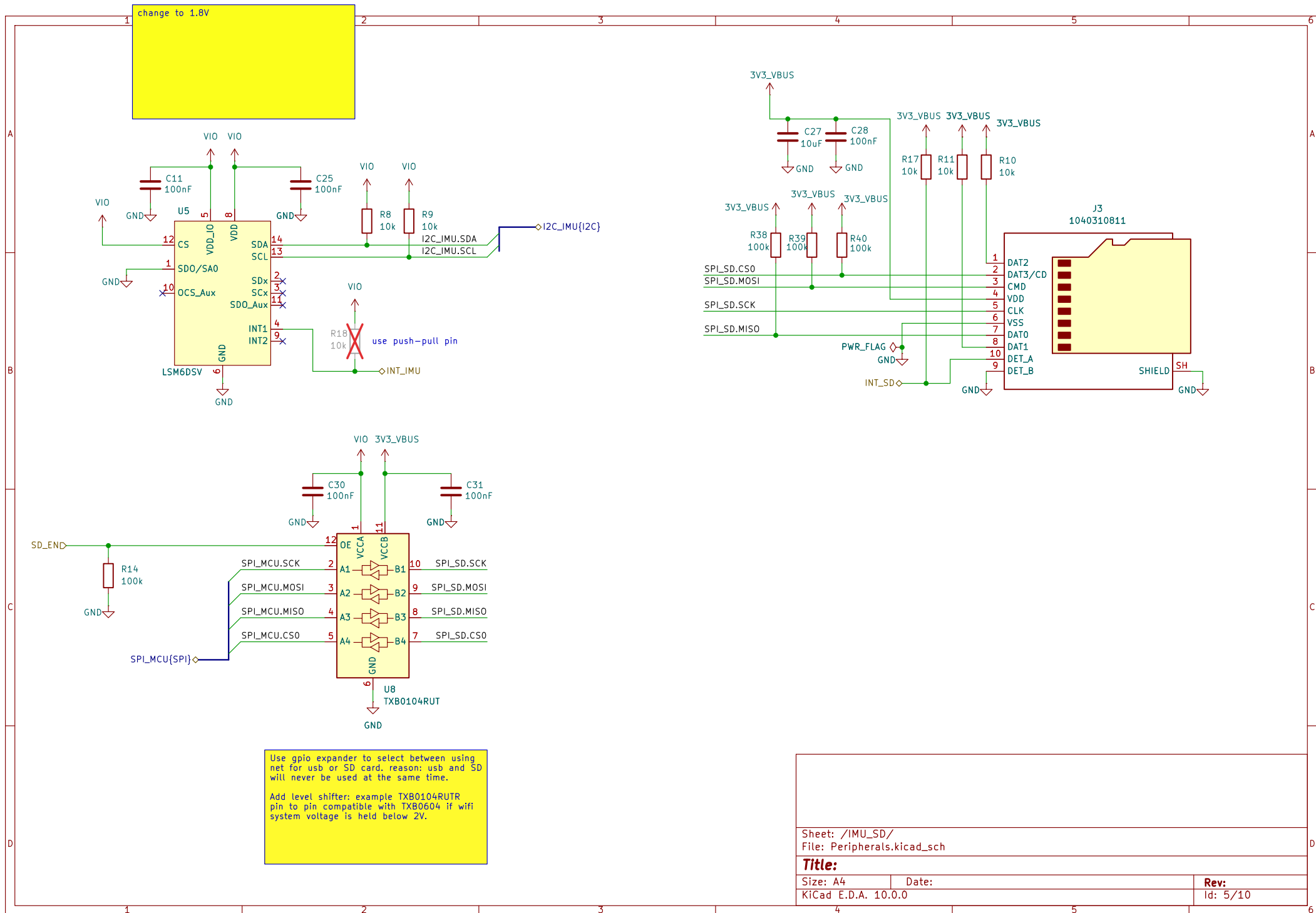
KiCad E.D.A. 10.0.0

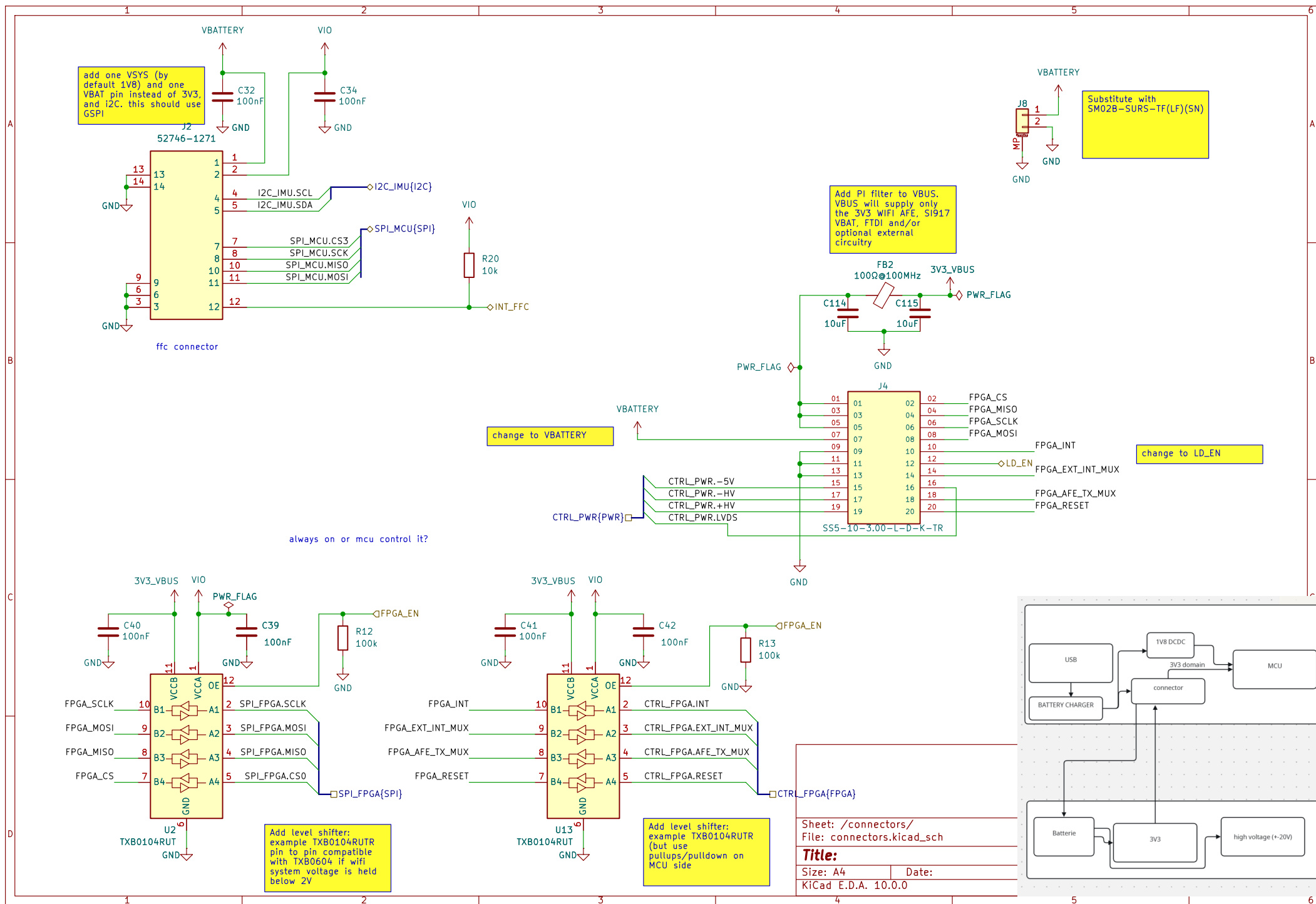
Rev:

Id: 4/10







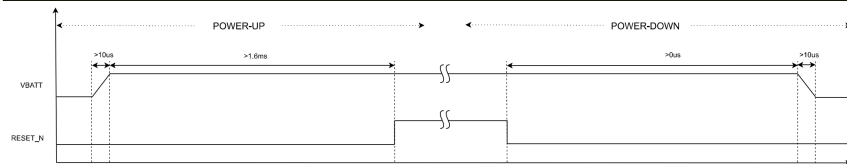


The IC generates a POC (Power On Control) signal – POC_OUT that is distributed to all I/O cells to prevent the I/O cells from powering up in an undesired configuration and is also used inside the IC to safe state the IC till a valid supply is available for proper operation. This power management is functional in both power up and power down sequences.

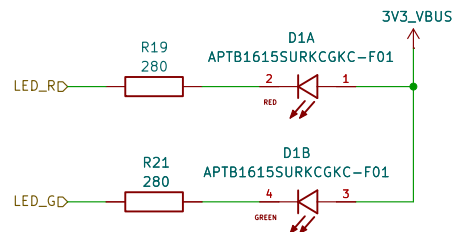
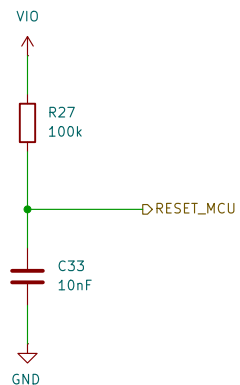
During power up, until the UULP_VBATT_1 and UULP_VBATT_2 (VBATT supply) reach 1.6V, the POC_OUT signal stays low. Once the VBATT supply exceeds 1.6V, the POC_OUT becomes high and normal operation of the IC starts.

Once the POC_OUT becomes high, it stays high. But if VBATT becomes lower than the Blackout threshold voltage, POC_OUT becomes low.

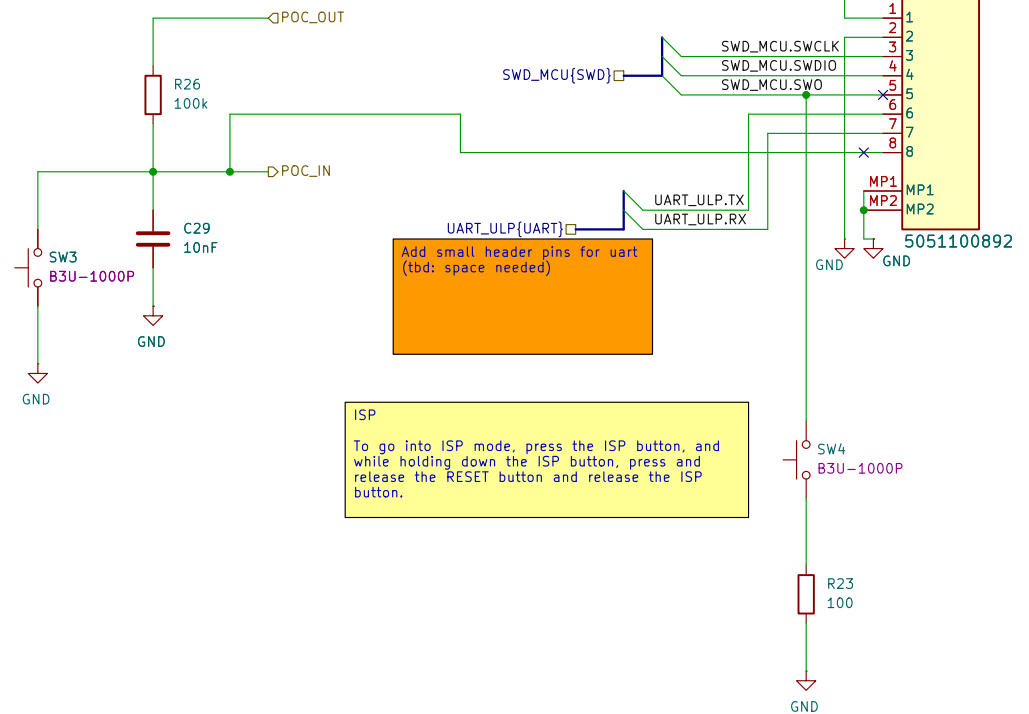
The following figure illustrates the power up sequence when POC_OUT is connected to POC_IN. As shown in the figure below, the RESET_N is high at least 1.6ms after VBATT supply is stable. The RESET_N signal can be controlled via options like an R/C circuit or another MCU's GPIO.



Add reset button?
already implemented in poc_in
poc_out configuratio with sw3.
poc_in is the master of the
reset.



Vdd = 3.3V
If = 5mA for both
Vf,red = 1.84V
Vf,green = 1.96V
--> R,red = 2920hm
R,green = 2680hm



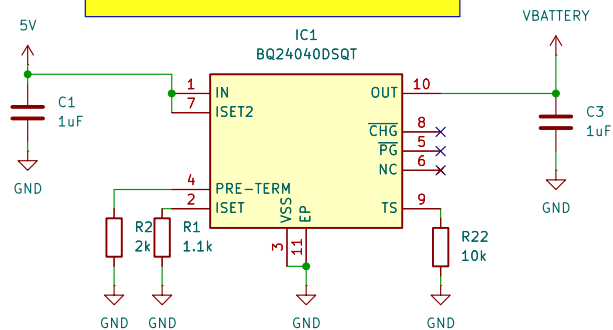
Sheet: /DEBUG, RESET & POC_IN, PROGRAMMING INTERFACE/
File: debug_interface.kicad_sch

Size: A4	Date:
KiCad E.D.A. 10.0.0	

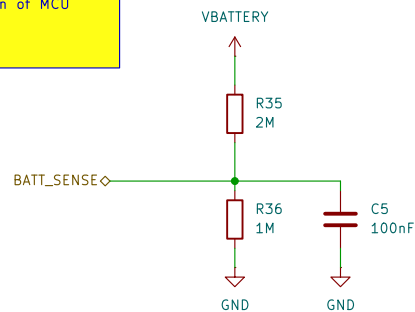
Date:

Rev:
Id: 8/10

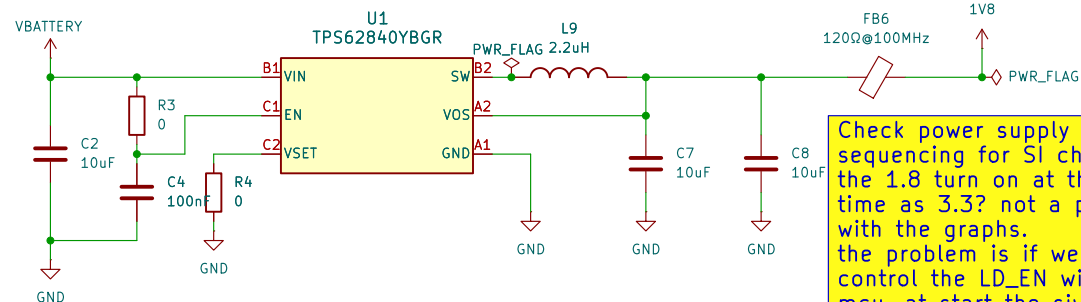
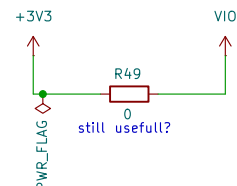
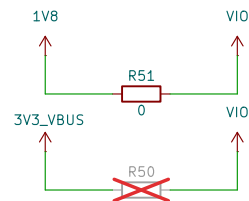
rail. the 4.2 voltage rail is then dc/dc converted to 1.8V (e.g. U1) to supply the entirety of the WIFI PCB



Instead of fuel gauge just add resistor divider to ADC pin of MCU



Make sure that only either R28 or R30 is mounted



Check power supply sequencing for SI chip: can the 1.8 turn on at the same time as 3.3? not a problem with the graphs. the problem is if we want to control the LD_EN with the mcu. at start the siwg can't work properly so can't use it. the VBATT of the mcu must be high with the other supply or the first one, but can be later

Power tree and configurations:

<https://t.ly/lrq2h>

Power scenario: resistors to place

All from battery, 1.8V: R12
All from battery, 3.3V: R21
VIO from TinyProbe: R19
MCU 3.3V only: R19, R21

Dev board (BRD4002A + BRD4338A) has around 20uF capacitance on VMCU line (VIO + 3.3V).

This board:
3.3V: 21.2uF
1.8V: 23.6uF

Sheet: /POWER CONVERSION/
File: power_conv.kicad_sch

Title:

Size: A4 Date:

KiCad E.D.A. 10.0.0

Rev:

Id: 10/10